ANALYSIS REPORT

TE Curve Verification Pipeline Selected Active Model Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the currently selected active model families against the repository held-out TE-curve test split. The `global` surface is intentionally excluded from this reduced decision report.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\polished_dataset` ;
- comparison mode: `selected_active_model_matrix` ;
- candidate count: `7` ;
- held-out curve count before candidate filtering: `100` ;
- percentage-error denominator: `peak_to_peak_truth` ;
- evaluated surface: `Fw` ;
- evaluated direction: `forward` ;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
`polished_setpoints_feedforward_Fw`	`polished_setpoints_model_archive`	`feedforward`	`Fw`	`forward`
`polished_setpoints_tree_Fw`	`polished_setpoints_model_archive`	`tree`	`Fw`	`forward`
`polished_setpoints_harmonic_regression_Fw`	`polished_setpoints_model_archive`	`harmonic_regression`	`Fw`	`forward`
`polished_setpoints_periodic_mlp_harmonic_Fw`	`polished_setpoints_model_archive`	`periodic_mlp_harmonic`	`Fw`	`forward`
`polished_setpoints_periodic_gru_sequence_Fw`	`polished_setpoints_model_archive`	`periodic_gru_sequence`	`Fw`	`forward`
`polished_setpoints_wave4_1_mae_robust_loss_Fw`	`polished_setpoints_model_archive`	`wave4_1_mae_robust_loss`	`Fw`	`forward`
`polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw`	`polished_setpoints_model_archive`	`wave4_2_quantile_p10_p50_p90`	`Fw`	`forward`

Exact Model Paths

Candidate	Family	Surface	ONNX Model Path	Python Model Path
`polished_setpoints_feedforward_Fw`	`feedforward`	`Fw`	`models/polished_dataset/setpoints/exported/feedforward/forward/2026-07-07-17-25-58__te_feedforward_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/feedforward/forward/2026-07-07-17-25-58__te_feedforward_fw__polished_setpoints/python/feedforward-epoch=042-val_mae=0.00168289.ckpt`
`polished_setpoints_tree_Fw`	`tree`	`Fw`	`models/polished_dataset/setpoints/exported/tree/forward/2026-07-07-09-34-40__te_tree_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/tree/forward/2026-07-07-09-34-40__te_tree_fw__polished_setpoints/python/tree_model.pkl`
`polished_setpoints_harmonic_regression_Fw`	`harmonic_regression`	`Fw`	`models/polished_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-23-29-07__te_harmonic_regression_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-23-29-07__te_harmonic_regression_fw__polished_setpoints/python/harmonic_regression-epoch=032-val_mae=0.01714993.ckpt`
`polished_setpoints_periodic_mlp_harmonic_Fw`	`periodic_mlp_harmonic`	`Fw`	`models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-02-15-59__te_periodic_mlp_harmonic_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/periodic_mlp_harmonic/forward/2026-07-08-02-15-59__te_periodic_mlp_harmonic_fw__polished_setpoints/python/periodic_mlp-epoch=051-val_mae=0.00120808.ckpt`
`polished_setpoints_periodic_gru_sequence_Fw`	`periodic_gru_sequence`	`Fw`	`models/polished_dataset/setpoints/exported/periodic_gru_sequence/forward/2026-07-08-22-57-44__te_periodic_gru_sequence_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/periodic_gru_sequence/forward/2026-07-08-22-57-44__te_periodic_gru_sequence_fw__polished_setpoints/python/periodic_gru_sequence-epoch=091-val_mae=0.00183161.ckpt`
`polished_setpoints_wave4_1_mae_robust_loss_Fw`	`wave4_1_mae_robust_loss`	`Fw`	`models/polished_dataset/setpoints/exported/wave4_1_mae_robust_loss/forward/2026-07-12-16-19-34__te_wave4_1_mae_robust_loss_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/wave4_1_mae_robust_loss/forward/2026-07-12-16-19-34__te_wave4_1_mae_robust_loss_fw__polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=105-val_mae=0.00179190.ckpt`
`polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw`	`wave4_2_quantile_p10_p50_p90`	`Fw`	`models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-13-15-27__te_wave4_2_quantile_p10_p50_p90_fw__polished_setpoints/onnx/model.onnx`	`models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-13-15-27__te_wave4_2_quantile_p10_p50_p90_fw__polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=157-val_mae=0.00180121.ckpt`

Metric Ranking

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
1	`polished_setpoints_wave4_1_mae_robust_loss_Fw`	0.016970	0.017294	36.104	88.789
2	`polished_setpoints_harmonic_regression_Fw`	0.017019	0.017299	38.005	77.612
3	`polished_setpoints_tree_Fw`	0.035091	0.035237	75.422	148.017
4	`polished_setpoints_feedforward_Fw`	0.036872	0.037171	79.766	139.388
5	`polished_setpoints_periodic_gru_sequence_Fw`	0.038470	0.038781	82.792	150.753
6	`polished_setpoints_periodic_mlp_harmonic_Fw`	0.046156	0.046484	99.650	180.552
7	`polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw`	0.051176	0.051284	110.194	191.941

Direction Breakdown

Direction	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
forward	polished_setpoints_wave4_1_mae_robust_loss_Fw	0.016970	0.017294	36.104	88.789
forward	polished_setpoints_harmonic_regression_Fw	0.017019	0.017299	38.005	77.612
forward	polished_setpoints_tree_Fw	0.035091	0.035237	75.422	148.017
forward	polished_setpoints_feedforward_Fw	0.036872	0.037171	79.766	139.388
forward	polished_setpoints_periodic_gru_sequence_Fw	0.038470	0.038781	82.792	150.753
forward	polished_setpoints_periodic_mlp_harmonic_Fw	0.046156	0.046484	99.650	180.552
forward	polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw	0.051176	0.051284	110.194	191.941

Curve Evidence

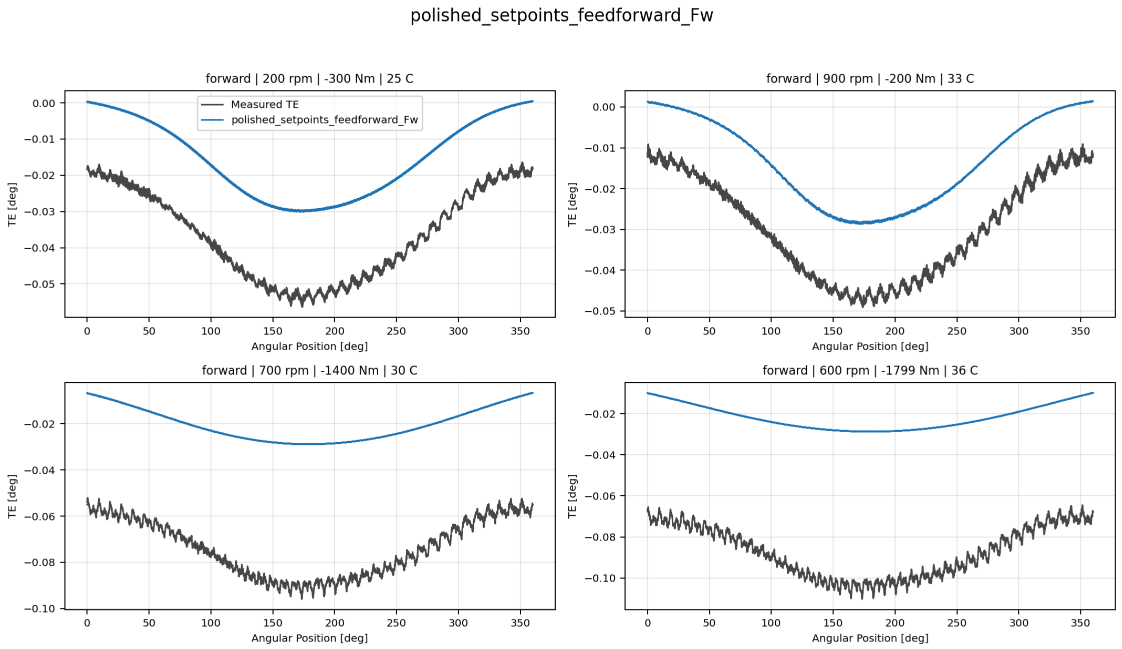
Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

Shared operating conditions

- 200 rpm, 300 Nm, 25 C
- 900 rpm, 200 Nm, 30 C
- 700 rpm, 1400 Nm, 30 C
- 600 rpm, 1800 Nm, 35 C

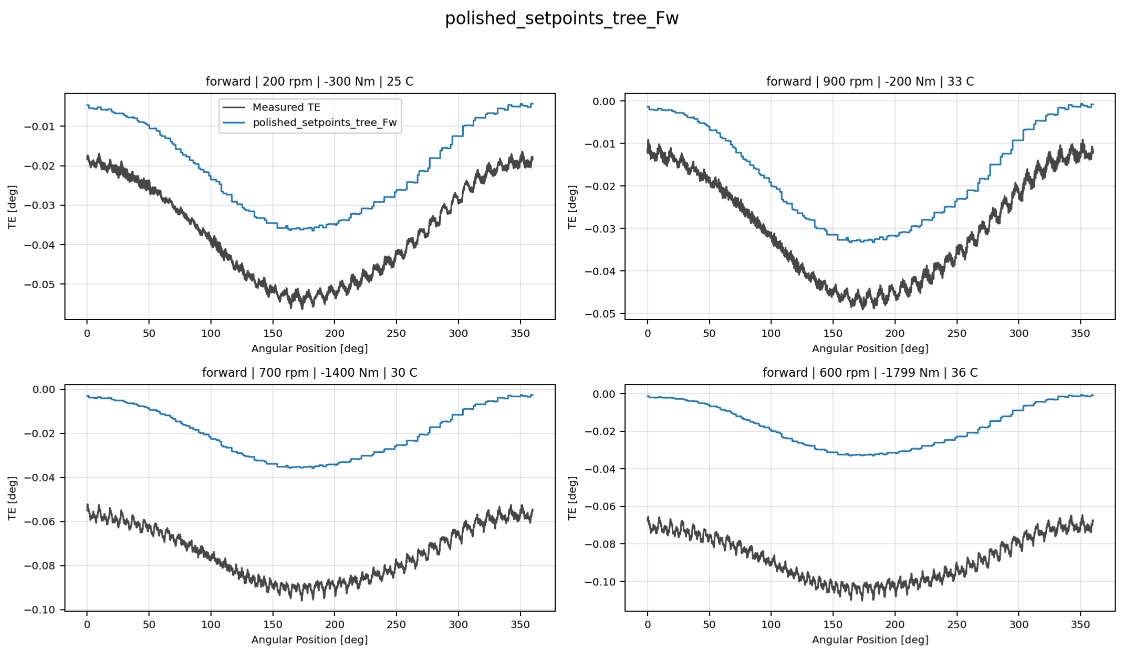
Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_setpoints_feedforward_Fw	Selected candidate	0.036872	0.037171	79.766	139.388
polished_setpoints_tree_Fw	Selected candidate	0.035091	0.035237	75.422	148.017
polished_setpoints_harmonic_regression_Fw	Selected candidate	0.017019	0.017299	38.005	77.612
polished_setpoints_periodic_mlp_harmonic_Fw	Selected candidate	0.046156	0.046484	99.650	180.552
polished_setpoints_periodic_gru_sequence_Fw	Selected candidate	0.038470	0.038781	82.792	150.753
polished_setpoints_wave4_1_mae_robust_loss_Fw	Surface leader	0.016970	0.017294	36.104	88.789
polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw	Selected candidate	0.051176	0.051284	110.194	191.941

polished_setpoints_feedforward_Fw



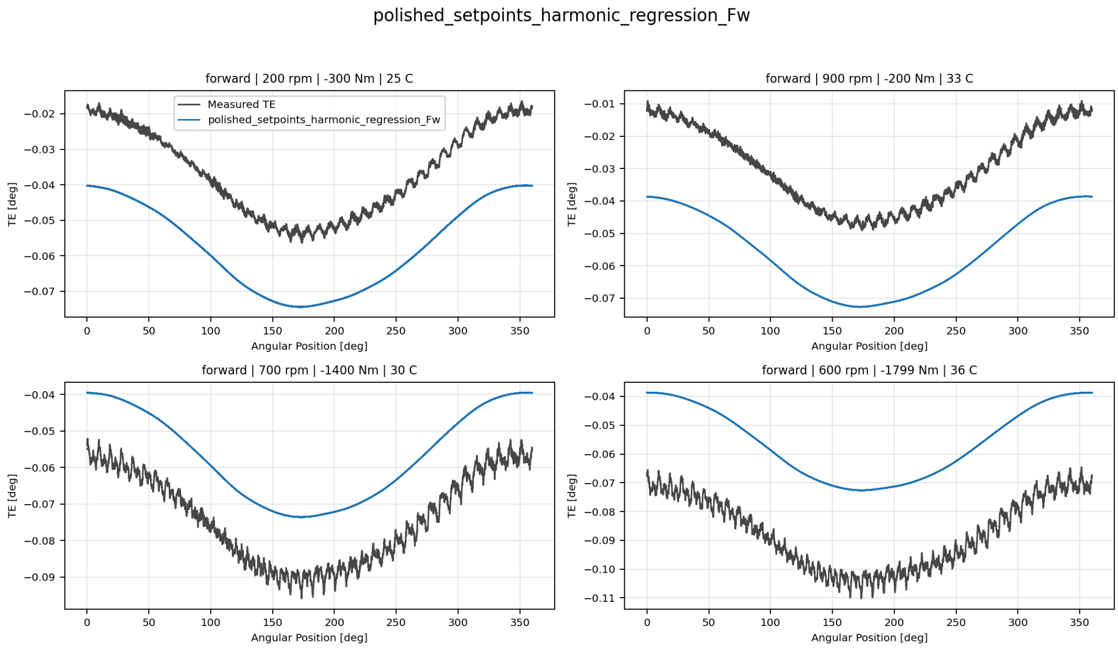
polished_setpoints_feedforward_Fw measured-versus-predicted TE collage

polished_setpoints_tree_Fw



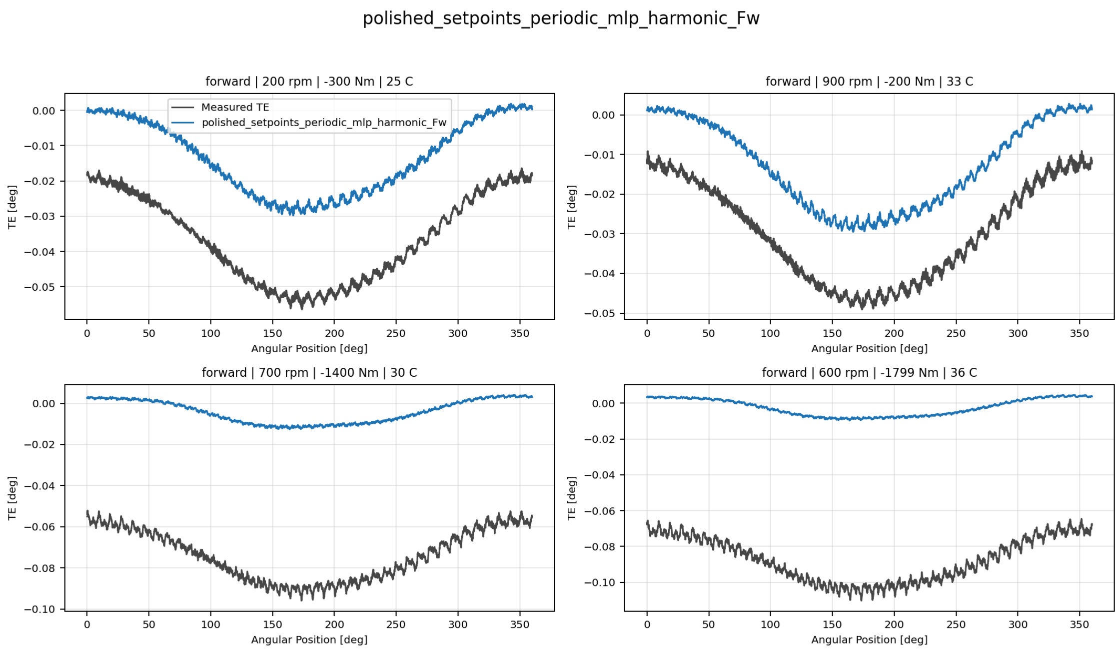
polished_setpoints_tree_Fw measured-versus-predicted TE collage

polished_setpoints_harmonic_regression_Fw



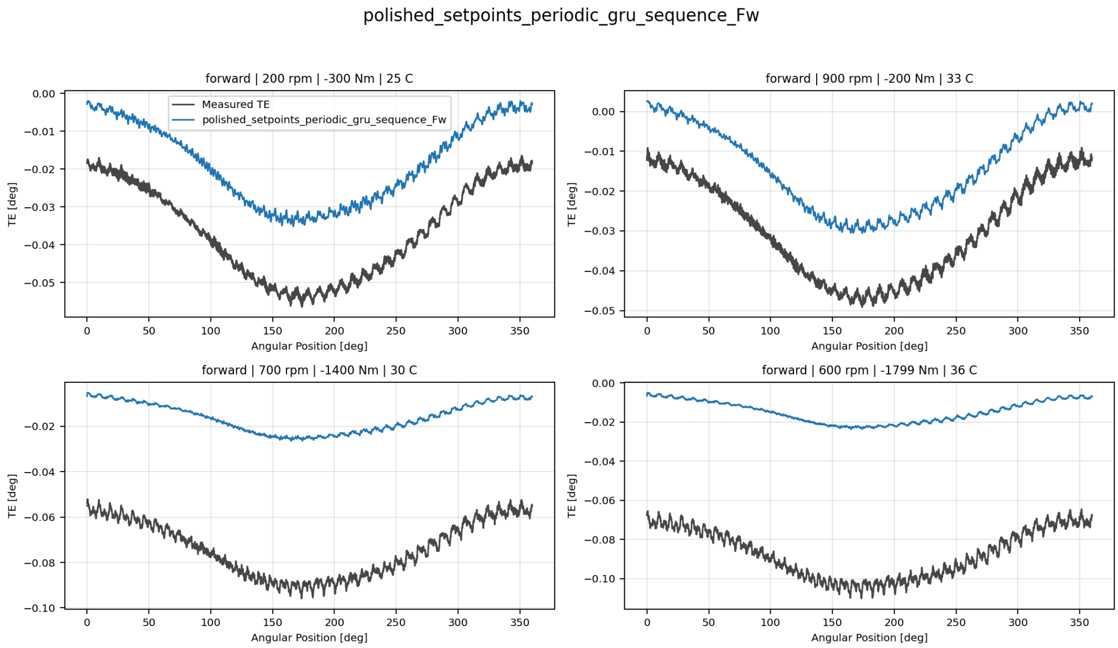
polished_setpoints_harmonic_regression_Fw measured-versus-predicted TE collage

polished_setpoints_periodic_mlp_harmonic_Fw



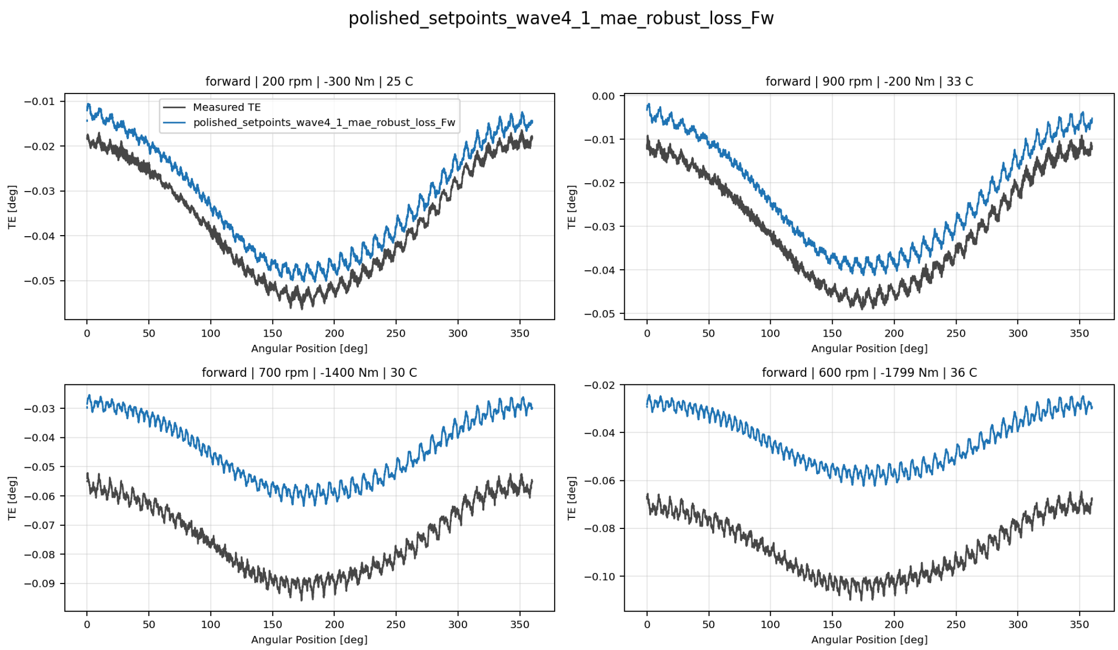
polished_setpoints_periodic_mlp_harmonic_Fw measured-versus-predicted TE collage

polished_setpoints_periodic_gru_sequence_Fw



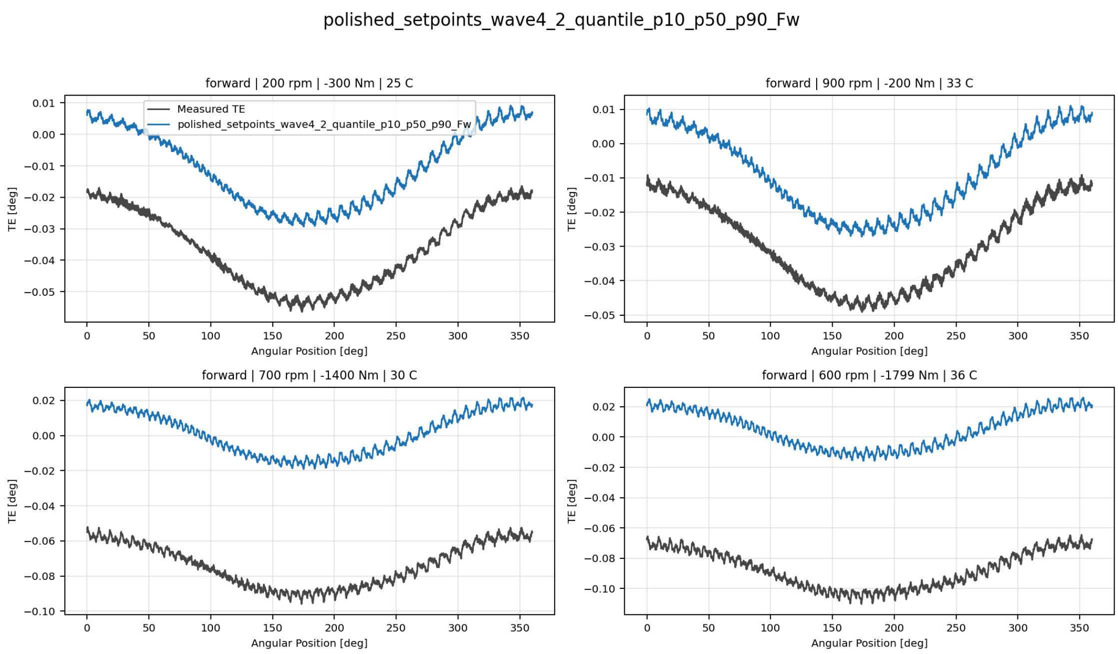
polished_setpoints_periodic_gru_sequence_Fw measured-versus-predicted TE collage

polished_setpoints_wave4_1_mae_robust_loss_Fw



polished_setpoints_wave4_1_mae_robust_loss_Fw measured-versus-predicted TE collage

polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw



polished_setpoints_wave4_2_quantile_p10_p50_p90_Fw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-19-17-42-20__track2_selected_active_polished_setpoints_matrix_track2_selected_active_polished_setpoints_forward_2026_07_19/validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-19-17-42-20__track2_selected_active_polished_setpoints_matrix_track2_selected_active_polished_setpoints_forward_2026_07_19\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\selected_active_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within this selected-active surface report. The curve-evidence section uses shared deterministic held-out operating conditions so visual shape fidelity can be compared across the selected families.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- RCIM Model-Bank Reproduction remains a separate paper-reference benchmark path and is not part of this selected-active model-only report.